INSTITUT FOR MATEMATIK
AARHUS UNIVERSITET

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December 5, 2025

Statistical Models
Information about the Exam

Modalities

Time and Place

Thursday, 22. January 2026, Room D02 (1531-015), starting 9:00

Each exam is supposed to take 25 minutes *including grading*.

Sequence of exams has been determined by the administration, and you can calculate your time approximately from your position in the list. Since it can happen that one of your fellow student cannot show up, please be there well in advance. We will hold a couple of small breaks in between though, and a lunch break.

Procedure

You will start with a small presentation (max 9-10 minutes) of an exam topics randomly drawn from the list below. If you have gotten approved all six assignments, you have the right to remove one topic before you draw.

In the second part, we will discuss questions posed by the censor or me, elaborating on the presentation and on other course subjects, also subjects not covered by presentation topics.

Notes or other aids

The exam is without aids, so: no notes. However, you are allowed to bring along one sheet with key words for your presentation and look at it once during the exam, to make sure that you did not forget anything you wanted to say :-). For example, you can start the presentation with writing a short list of main topics you want to address on the blackboard.

List of Presentation Topics

1. Likelihood theory
2. Exponential distribution families
3. Generalized linear models: general principles and results
4. GLM for count data or GLM for binary data and proportions
5. Bayesian statistics: idea and summarizing the posterior distribution
6. Bayesian statistics: choices for the prior distribution
7. Computational methods for Bayesian statistics, MCMC
8. Bayesian model selection and assessment

These topics are too broad to be presented comprehensively in 9 minutes. Focus on material you deem most interesting and relevant.

Have a nice Christmas time, and good succes with your exam preparation :-)

Ute